MIL-D-26392B (USAF)
30 September 1980
SUPERSEDING
MIL-D-26392A (USAF)
13 September 1963

MILITARY SPECIFICATION

DUMMY CONVERTER, LIQUID OXYGEN INDICATOR SYSTEM 10 LITER CRU-23/A

This specification is approved for use by The Department of The Air Force, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of dummy converter (electrical capacitor) having a capacitance equivalent to an empty, 10-liter, liquid oxygen converter, with respect to a capacitance type quantity gaging system. The dummy converter is designated CRU-23/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of document. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-packaging, Methods Of
MIL-C-5501	Cap and Plug, Protective, Dust And Moisture Seal.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-25516	Connector, Electrical, Miniature, coaxial, Environment Resisting Type, General Specification for

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-721	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety.
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 First article inspection. This specification makes provisions for First Article Inspection. (see 4.4).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4 Design and construction. The design and construction of the dummy converter shall be in accordance with Figure 1.

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The complete dummy converter shall consist of the following components:

Capacitor
Electrical Connectors
Case

3.4.1 Capacitor. The capacitor shall have an electrical capacitance of 123.5 PF, plus or minus 1.0 PF. The rated voltage of the capacitor shall be 110 volts.

3.4.2 Electrical connectors. The electrical connectors shall be as shown on figure 1, and they shall be of the Jack Body Type.

3.4.3 Case. The case shall conform to the configuration and dimensions show on figure 1. The case shall provide a moisture tight seal for the capacitor. The case shall be made of corrosion-resistant material or from material that has been suitably protected against corrosion. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.5 PERFORMANCE

3.5.1 Capacitance at room temperature. The capacitance at room temperature shall be 123.5 Picofarads (PF, plus or minus 1.0 PF).

3.5.2 Capacitance at low temperature. The capacitance at minus 65 degrees Fahrenheit shall remain within the tolerance specified for room temperature.

3.5.3 Capacitance at high temperature. The capacitance at plus 260 degrees Fahrenheit shall remain within the limits specified for room temperature.

3.5.4 Immersion (leakage). The case shall not leak when the dummy converter is subjected to the immersion (leakage) test.

3.5.5 Reliability. The dummy converter shall have a specified mean time between failures (MTBF) of not less than 300 hours at the 90 per cent lower confidence limit. (This is equivalent to minimum acceptable MTBF of 200 hours, and to a reliabilty of .9933 for 2-hour mission).

3.5.6 Maintainability. No corrective maintenance task time, as defined in MIL-STD-721, shall exceed one hour of maintenance downtime.

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3.6 Weight. The complete dummy converter shall weigh not more than 0.25 pound.

3.7 Identification of product. The dummy converters shall be marked for identification in accordance with MIL-STD-130 and shall include the following information:

Dummy Converter
10 Liter
MIL-D-26392B
Type CRU-23/A
Manufacturer's Part Number *
Contract Number *
Manufacturer's Name or Trademark *

*The manufacturer shall include applicable information.

3.8 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-STD-100.

3.9 Workmanship. The dummy converters shall be uniform in quality and shall be free from irregularities or defects which could adversely affect safety, performance, reliability, or durability. The converters shall be free of oil, grease, fuel, water, dust, dirt, or any other foreign matter.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the dummy converter shall be classified as follows:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 Inspection conditions. Unless otherwise specified, all tests required by this specification shall be made at an

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atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 degrees, plus or minus 18 degrees Fahrenheit, and at a relative humidity of 80 percent or less. Where tests are conducted with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading.

4.4 First article inspection. First article inspection shall consist of all the examinations and tests described under test methods, paragraph 4.6.

4.4.1 Samples. The first article inspection shall be performed on one dummy converter which is representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production from contract to contract, submission of further first article samples or test reports may be waived at the discretion of the procuring activity. Approval of the first article inspection or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection.

4.4.2 Manufacturer's inspection report. A test report prepared in accordance with MIL-STD-831 shall include the results of all tests including a detailed statement indication compliance or extent of noncompliance with each requirement of this specification. The report shall be forwarded to the procuring activity together with the tested samples unless otherwise specified.

4.4.3 Accept-reject criteria for reliability testing. The accept-reject criteria for reliability testing of the first articles shall be in accordance with Test Plan I of MIL-STD-781.

4.5 Quality conformance inspections. Quality conformance inspections shall consist of:

(a) Individual inspections.....(4.5.1).
(b) Sampling inspections.....(4.5.2).

4.5.1 Individual inspections. Each dummy converter on the contract or order shall be subjected to the following examination and tests. Any dummy converter containing a defect shall be rejected.

(a) Examination of product (4.6.1).
(b) Capacitance at room temperature(4.6.2).

4.5.2 Sample inspection. Two dummy converters shall be

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selected at random from each lot of 200 or fraction thereof and subjected to the following tests as described under 4.6:

a. Capacitance at low temperature (4.6.3).
b. Capacitance at high temperature (4.6.4).
c. Immerson (leakage) (4.6.5).

4.5.2.1 Rejection and retest. If one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld unitl the extent and cause of the failure have been determined. The contractor shall explain fully to the Government representative the cause of the failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.6 TEST METHODS

4.6.1 Examination of product. Each dummy converter shall be carefully examined to determine conformance to the requirements of this specification with respect to material, design, construction, weight, marking, and workmanship.

4.6.2 Capacitance at room temperature. The capacitance shall be determined at room temperature by using a master test instrument of suitable accuracy. The capacitance shall be 123.5 PF, plus or minus 1.0 PF.

4.6.3 Capacitance at low temperature. The dummy converter shall be cold soaked at minus 65 degrees Fahrenheit for at least 8 hours. While still at this temperature, the capacitance shall be determined with a suitable master test instrument. The capacitance shall be 123.5 PF, plus or minus 1.0 PF.

4.6.4 Capacitance at high temperature. The dummy converter shall be maintained at a temperature of at least plus 260 degrees Fahrenheit for at least 8 hours. While still at this temperature, the capacitance shall be determined with a suitable master test instrument. The capacitance shall be 123.5 PF, plus or minus 1.0 PF.

4.6.5 Immersion (leakage). The dummy converter shall be subjected to the immersion (leakage) test specified in method 512, procedure I of MIL-STD-810.

4.6.6 Reliability test level. The reliability test level shall consist of temperature and vibration cycling as follows: The dummy converter shall be stabilized at a temperature of minus 65 degrees Fahrenheit, then stabilized at a temperature of plus 260 degrees Fahrenheit and held for at least two hours, then returned to a temperature of minus 65 degrees Fahrenheit. The

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transition period in changing between temperature extremes shall not exceed 40 minutes.The dummy converter shall be subjected to at least four of these temperature cycles per 24 hours. The dummy converter shall be subjected vibration at any frequency between 20 and 60 cycles per second and an applied acceleration of plus or minus 2g for at least 10 minutes within each hour of this test. Rated voltage plus 0 minus 2 percent shall be applied continuously during this test. A failure shall be identified as any malfunction which causes excessive error in the capacitance value of the dummy converter. The dummy converter shall be checked at least once every 24 hours at each temperature extreme to determine if a failure has occurred. No parts shall be replaced as preventive maintenance during reliability testing.

4.6.7 Maintainability verification. Time studies of all corrective maintenance tasks for all expected failure modes shall be conducted by simulating failures. Timing of each maintenance task shall be performed at least three times to determine average values. The maintenance tasks considered shall be only those related to field level maintenance. The time to accomplish each maintenance task shall include time for preparation, recognition, fault location, removal (based only on the installation provisions of the item), fault correction, adjustment and realignment, final test, and replacement, but shall exclude supply time and administrative time. In the case of repetitive subtasks which occur by the nature of the sequence of maintenance actions (e.g., removal), it shall not be necessary to redetermine the average time required.

4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-STD-794, Level A or C, as specified by the procuring activity.

5.1.1 Level A. The dummy converter shall be preserved and packaged in quantity unit packages of one each in accordance with Method 1A-8 of MIL-P-116. Completed packages to be subjected to the free fall rough handling drop tests of MIL-STD-794.

Electrical connectors shall be protected with caps or plugs conforming to MIL-C-5501.

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5.1.2 Level C. Same as Level A except Method III of MIL-P-116 applies in lieu of Method 1A-8.

5.2 Packing.

5.2.1 Level A. Dummy converter, packaged as specified in paragraph 5.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity.

5.2.2 Level B. Dummy converter packaged as specified in paragraph 5.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636 unless otherwise specified by the procuring activity.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for installation or immediate use. This level shall conform to the applicable carrier rules and regulations and may be the contractor's commercial practice, provided the latter meets the requirements of this level.

5.3 Marking. The exterior and interior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear on each unit package and displayed on at least two sides of the container. The color of the labels will be a conspicuous green.



6. NOTES

6.1 Intended use. The dummy converter covered by this specification is intended to be used in F-111, F-14, and F-15 aircraft when one of the converters is not installed.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

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b. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
c. Items of data required (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officer's any of the data specified in applicable documents listed in Section 2 of this specification, or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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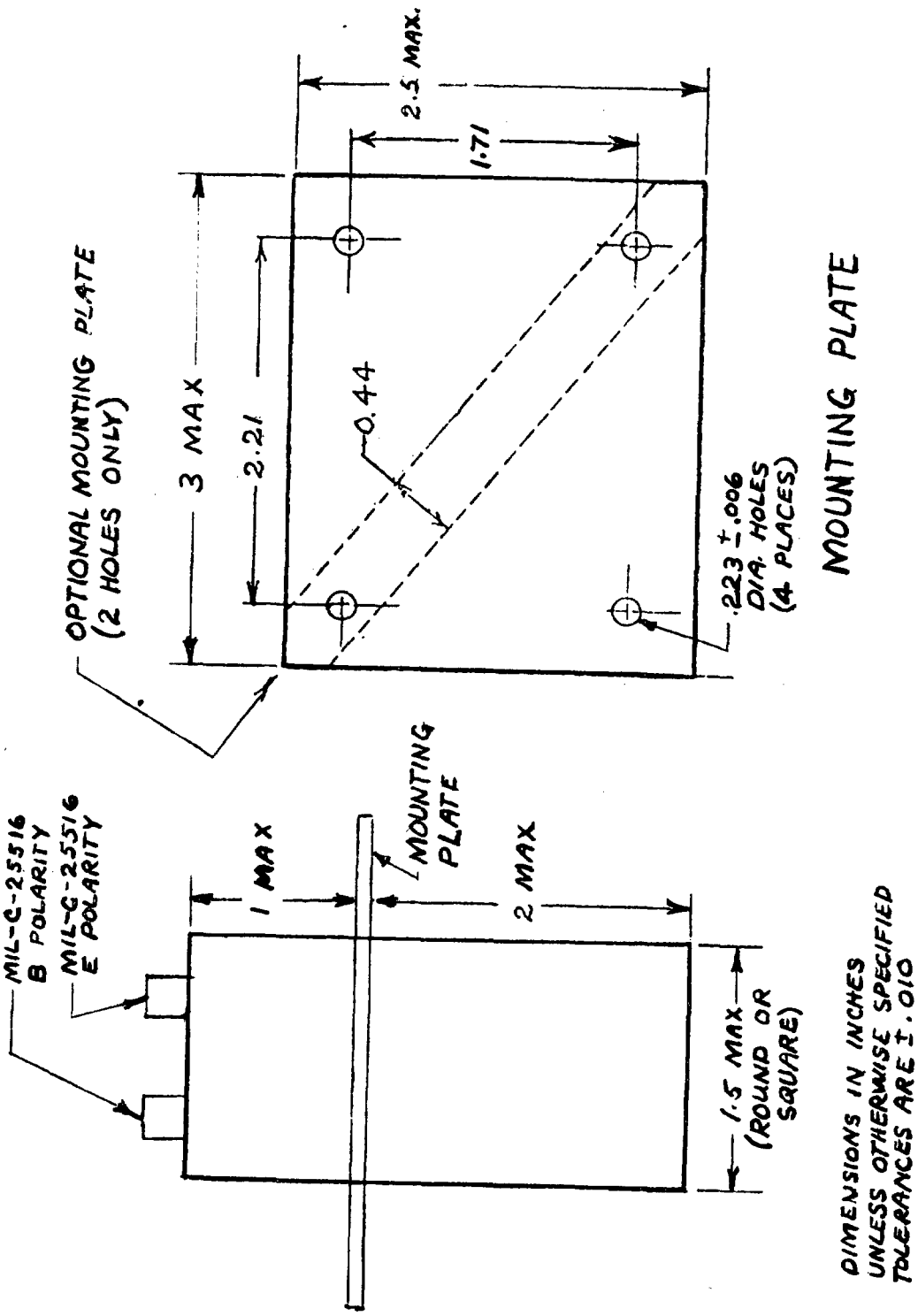


FIGURE 1 - Electrical connector and mounting plate

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